PROJECT DOCUMENTATION

Database Backup & Recovery Plan

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1. Purpose

This Database Backup and Recovery Plan aims to:

- **Ensure Data Integrity:** Maintain accurate, consistent, and reliable data even after restoration activities.
- **Support Business Continuity:** Minimize downtime and operational impact during data loss events, system failures, or cyber threats.
- **Meet Regulatory Requirements:** Align organization's data protection practices with legal and industry standards to prevent compliance risks.
- **Build Stakeholder Confidence:** Provide assurance to customers, employees, and stakeholders that company reliably safeguards its data.

2. Scope

This plan applies to all organizational databases and covers:

- **All Database Platforms:** SQL Server, Oracle, PostgreSQL, MongoDB, others.
- **All Departments Using Database:** Including HR, Finance, Operations, and Customer Support.
- **On-Premises, Off-site and Cloud Databases:** Whether hosted in the primary data center, disaster recovery site, or cloud infrastructure.
- **Archived and Legacy Data:** Ensures proper backup and restoration for historical or compliance-related datasets.

3. Roles and Responsibilities

3.1 Database Administrator (DBAs)

- Execute and monitor scheduled backups.
- Lead data restoration efforts and ensure timely recovery.
- Perform routine tests for backup integrity and recovery performance.
- Continuously refine the backup and recovery processes.

3.2 IT Managers

- Allocate necessary hardware, software, and personnel for backup operations.
- Manage budgeting for tools, infrastructure, and training.
- Provide skill-development opportunities for DBAs and IT staff.
- Maintain coordination among DBAs, security teams, and other departments.

3.3 IT Security Teams

- Ensure all backup data is securely encrypted on-site and off-site.
- Enforce strict access control for backup repositories.
- Assist DBAs during incidents involving data breaches or cyberattacks.
- Conduct periodic audits to identify and mitigate vulnerabilities.

3.4 Disaster Recovery Teams

- Provide necessary infrastructure when system or data center disruptions occur.
- Conduct semi-annual recovery drills, including database restoration tasks.
- Document drill results, lessons learned, and improvement areas.

3.5 Compliance Officers or Legal Teams

- Ensure adherence to regulatory and industry standards.
- Define and enforce legal retention periods for data.
- Support external or internal audits related to backups and recovery.

4. Backup Strategy

The following table below outlines the various backup methodologies adopted by IT Corp, detailing their purpose, when they are executed, and how long the backup data is retained:

Strategy	Description	Schedule	Retention Period
Full Backups	Entire database copy	Weekly - Saturday 2:00 AM	30 days
Differential Backups	Data changed since last full backup	Daily at 2:00 AM, (except Saturday)	7 days
Transaction Log Backups	Captures logs for point-in-time recovery	Every 6 hours	48 hours
Backup Storage	**Primary:** On-site secure server **Secondary:** Off-site cloud storage	N/A	N/A

5. Recovery Procedure

The table below details the actions that will be taken in response to various database crises, ensuring a clear and prompt recovery process.

Scenario	Action
Minor Data Loss/Corruption	Restore latest backup; apply logs sequentially.
Major System or Data Center Outage	Restore from off-site cloud storage; activate DR procedures.
Database Server Failures	Use standby servers; begin restoration as required.

6. Security Measures

The table below encapsulates the safeguards put in place to protect our backup data, including encryption standards and access limitations.

Measure	Description
Encryption	All backups are protected using AES-256-bit encryption.
Access Control	Restrictive access granted only to authorized DBAs and IT security personnel.

7. Testing

7.1 Objective

Ensure backup and recovery processes perform reliably and meet business requirements.

Schedule

- **Frequency:** Twice a year.
- **Next Scheduled Test:** February 20, 2026.

Mock Recovery Drills

- Simulate various disaster scenarios to test the effectiveness of recovery procedures.
- Include DBAs, IT managers, and security teams.

Success Criteria

- **Recovery Time Objective (RTO):** Successfully restore operations within 4 hours of initiating the recovery procedure.
- **Recovery Point Objective (RPO):** Maximum allowed data loss is 30 minutes.

Post-Drill Review

- Conduct review sessions within 7 days of each drill.
- Document successes, challenges, and improvement points.
- Update procedures accordingly.

Training and Awareness

- Conduct refresher or skill-enhancement training after every drill.
- Inform all stakeholders of updated policies or procedure changes.

8. Documentation and Logs

8.1 Backup Logs

- **Purpose:** To maintain a comprehensive record of all backup activities.
- **Details Captured:** Date, time, database name, backup size, location, backup type (full, differential, transaction log), and any errors or warnings.
- **Storage:** Logs will be stored on a dedicated server with restricted access. They will be retained for 2 years.
- **Review:** Monthly reviews by DBAs to identify patterns, failures, or inefficiencies.

8.2 Recovery Logs

- **Purpose:** To document all recovery efforts and actions.
- **Details Captured:** Cause of data loss or system failure, date and time of the event, steps taken for recovery, recovery start and end time, and outcomes.
- **Storage:** Logs will be stored alongside backup logs on the dedicated server with restricted access and will be retained for 2 years.
- **Review:** After every recovery event, logs will be reviewed by a joint team of DBAs and IT managers to ascertain the effectiveness of the recovery and areas of improvement.

8.3 Annual Documentation Review

- Ensure policies align with new technologies and best practices.
- Conducted by senior DBAs, IT managers, and security teams.
- All updates must be recorded and communicated.

9. Review and Updates

Ensuring the relevance and efficiency of the Backup and Recovery Plan requires regular scrutiny and updates. As IT Corp's data infrastructure evolves, the plan must keep pace.

Review Parameter	Description	Frequency
Technological Changes	Update plan based on new database tools or systems.	As required
Data Growth	Adjust schedules or storage based on increased data volume.	Quarterly
Infrastructure Changes	Update procedures after major server or platform modifications.	Bi-annually
Security Protocols	Implement new security mechanisms to protect	Annually

	backups.	
Feedback from Drills	Integrate lessons learned from disaster recovery drills.	Post-drill

10. Conclusion:

The integrity, security, and accessibility of IT Corp's data are essential to the organization's operations. This Backup and Recovery Plan ensures that IT Corp can rapidly restore operations and minimize disruptions in the event of data loss, system failures, or catastrophic incidents.